Seaborn

• Seaborn is a **powerful Python data visualization library built on top of Matplotlib**. It provides a high-level interface for creating **attractive** and informative **statistical** graphics**.** Seaborn is particularly useful for visualizing complex datasets with multiple variables, making it a popular choice among data scientists and analysts**.**

Pip install seaborn

Provides default datasets for development and testing

```
[4]: import seaborn as sns

     datasets = sns.get_dataset_names()

     for d in datasets:
         print(d)

anagrams
anscombe
attention
brain_networks
car_crashes
diamonds
dots
dowjones
exercise
flights
fmri
geyser
glue
healthexp
iris
mpg
penguins
planets
seaice
taxis
tips
titanic
```

Example

Let us find out details about the tips paid and calculate the total tips paid taking the first 5 records, for testing purpose**.**

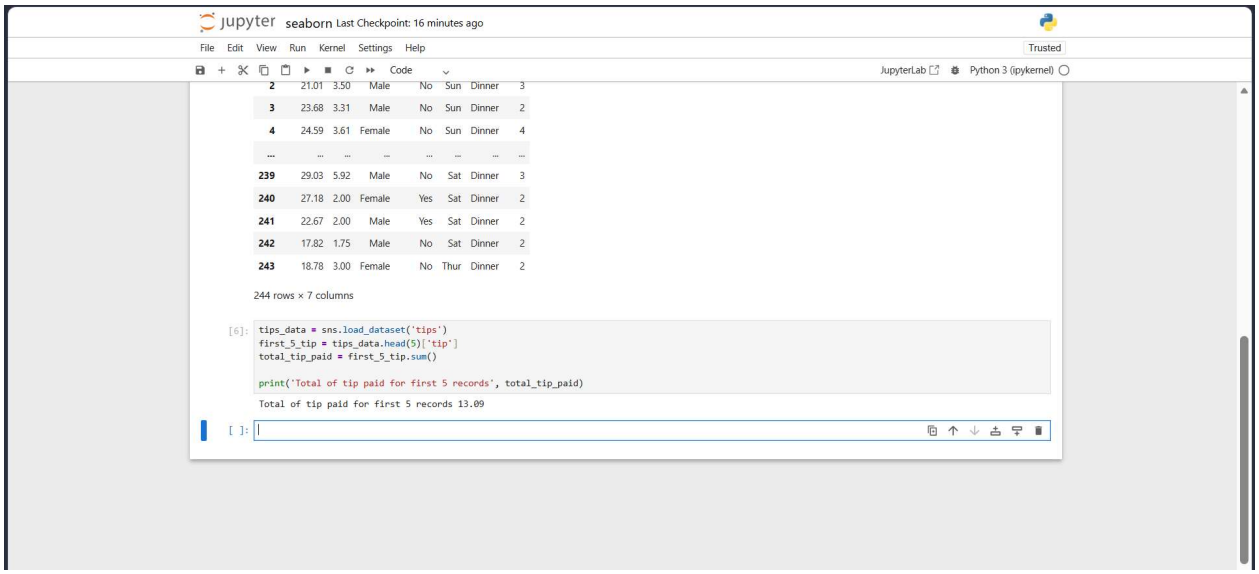
```
[5]: tips_data = sns.load_dataset('tips')
     tips_data
```

[5]:

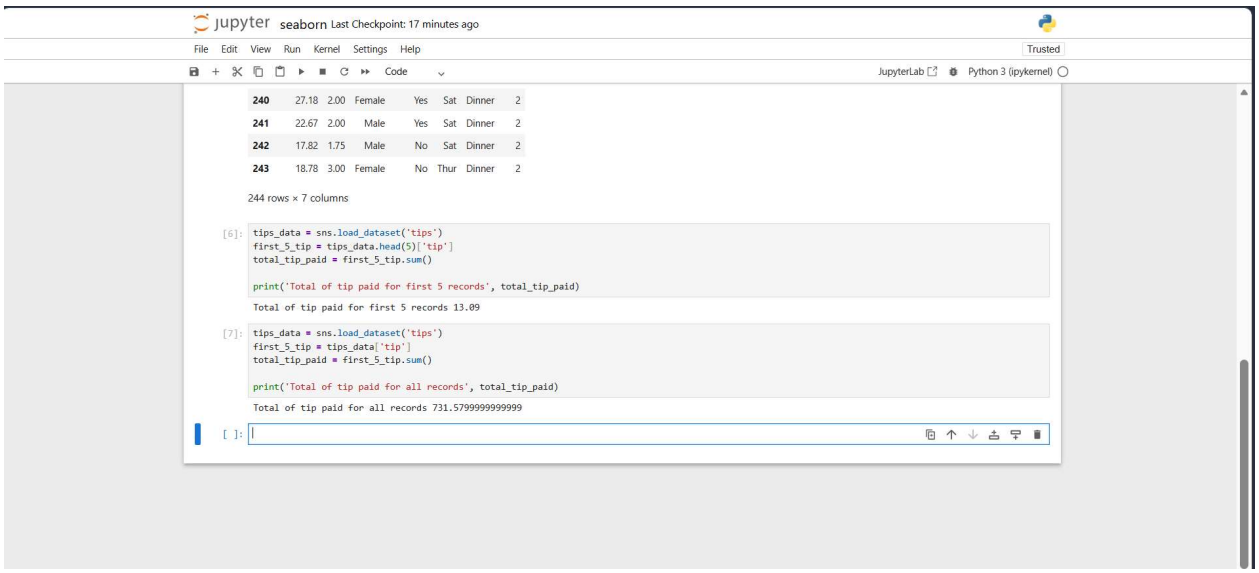
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

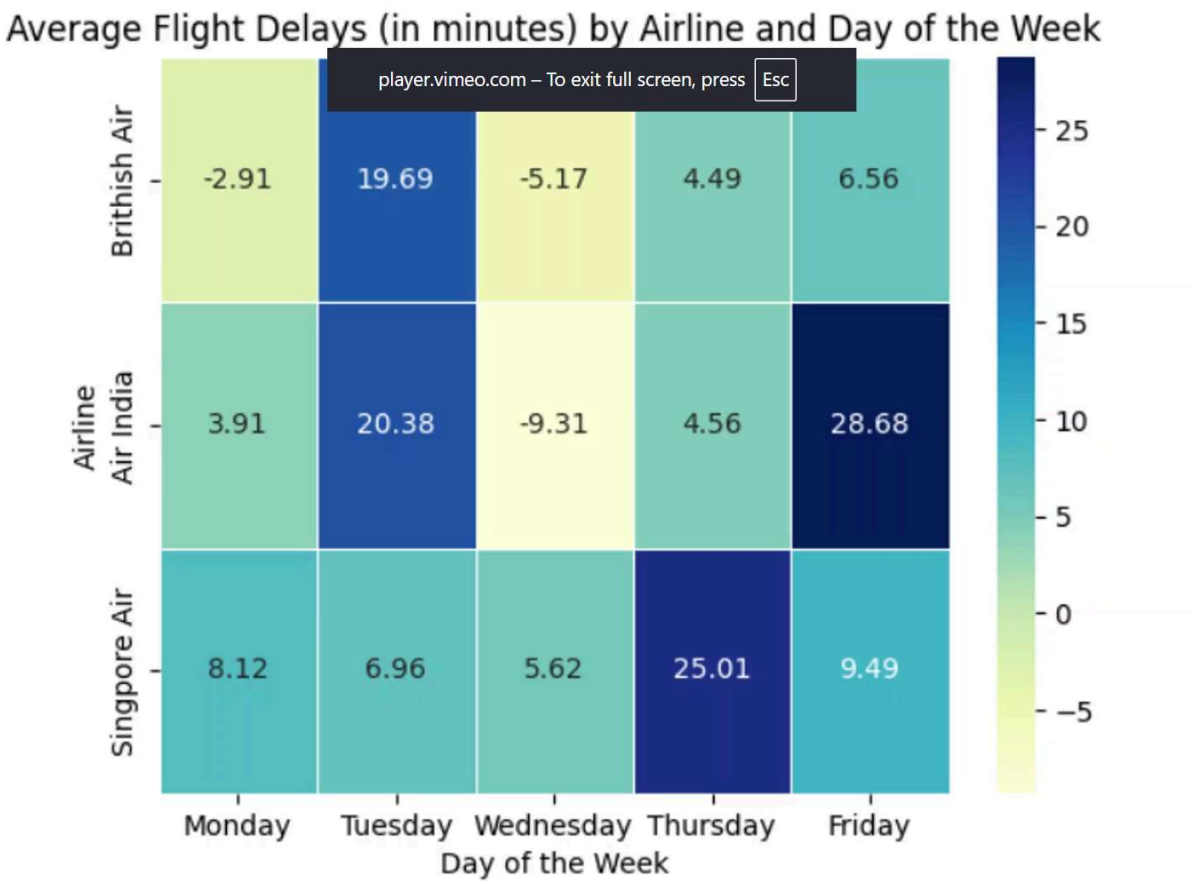
Total tips paid for first 5 records



Total tips paid for first all records



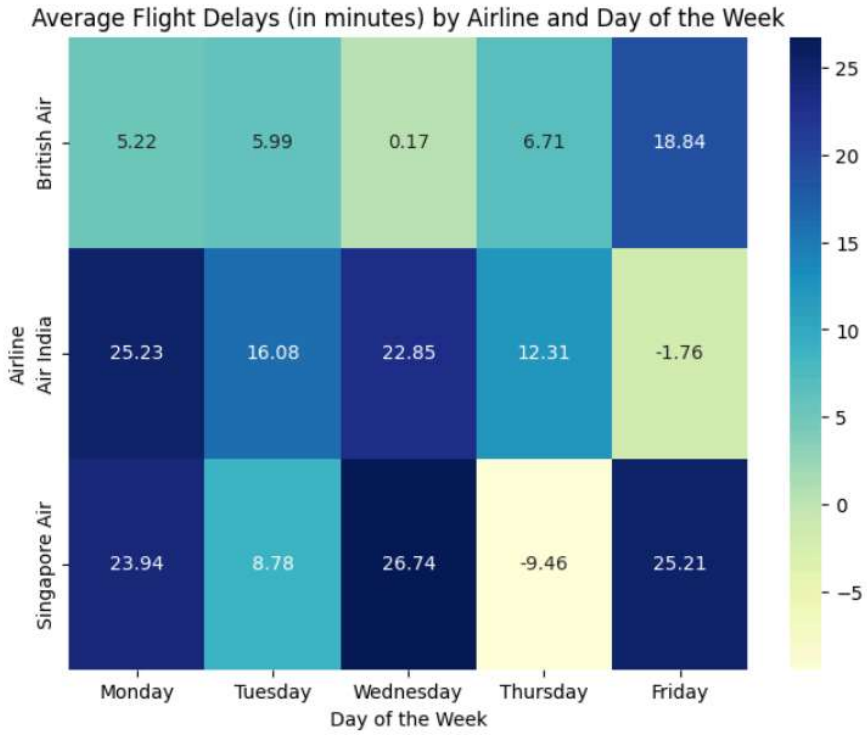
Heatmap:-



Example:-

```
[8]: import seaborn as sns
     import numpy as np
     import matplotlib.pyplot as plt

     # Labels
     airlines = ['British Air', 'Air India', 'Singapore Air']
     days_of_week = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday']

     # Generate random delay data
     delay_data = np.random.uniform(low=-10, high=30, size=(3, 5))

     # Create heatmap
     plt.figure(figsize=(8, 6))
     sns.heatmap(delay_data,
                 annot=True,
                 cmap='YlGnBu',
                 fmt=".2f",
                 xticklabels=days_of_week,
                 yticklabels=airlines)

     # Titles and labels
     plt.title('Average Flight Delays (in minutes) by Airline and Day of the Week')
     plt.xlabel('Day of the Week')
     plt.ylabel('Airline')

     # Show plot
     plt.show()
```

Output:-


Average Flight Delays (in minutes) by Airline and Day of the Week
British Air
Air India
Singapore Air
Airline
5.22
5.99
0.17
6.71
18.84
25.23
16.08
22.85
12.31
-1.76
23.94
8.78
26.74
-9.46
25.21
Monday
Tuesday
Wednesday
Thursday
Friday
Day of the Week
25
20
15
10
5
0
−5